

AutoInsight

South African Car Market Intelligence Platform

PROJECT ROADMAP & TEAM RESPONSIBILITIES

Detailed Timelines, Deliverables & Task Allocation

Project Duration	21 February 2026 – 1 June 2026 (14 weeks)
Weekly Checkpoints	Every Saturday
Team Size	5 members (2 Backend, 3 Frontend)
Backend Team	Damian, Jude
Frontend Team	Mpho, Taureen, Mpumelelo
Approach	Concurrent track development
Launch Date	1 June 2026

1. Executive Summary

This document will be the comprehensive project roadmap for AutoInsight, covering the 14-week development timeline from 21 February 2026 to 1 June 2026. The project will have a concurrent development model where frontend and backend teams work at the same time from Week 1, coming together at our integration milestones.

- Concurrent Development: Frontend and Backend teams start together in Week 1
- Sample Data Approach: Frontend uses mock data while Backend builds
- Weekly Checkpoints: Every Saturday for progress review and blocker resolution
- Clear Ownership: Each team member has defined feature areas and deliverables

1.1 Five-Layer Architecture

Layer 1: Data Scraping	Python script scrapes cars.co., AutoTrader, WeBuyCars (Backend)
Layer 2: MySQL Database	Database schema with dimension/fact tables (Backend)
Layer 3: Analytics & ML	Price prediction model and API endpoints (Backend)
Layer 4: Web Platform	React consumer-facing interface (Frontend)
Layer 5: Power BI Dashboard	Embedded visualisations via iFrame (Frontend + Backend)

2. High-Level Timeline: 14 Weeks

The project is divided into four phases with both teams working at the same time throughout.

Phase 1: Foundation (Weeks 1-3: 22 Feb - 14 Mar)

Week 1 (22-28 Feb): Backend: DB schema, scraper optimisation. Frontend: React setup, component library.

Week 2 (1-7 Mar): Backend: REST API design, authentication. Frontend: Core UI components, routing.

Week 3 (8-14 Mar): Backend: Database optimised, 70,000 vehicles. Frontend: Vehicle cards, search UI.

MILESTONE 1: Database populated, API documented, Frontend UI base completed

Phase 2: Core Development (Weeks 4-7: 15 Mar - 11 Apr)

Week 4 (15-21 Mar): Backend: ML feature engineering. Frontend: API integration (sample data to real).

Week 5 (22-28 Mar): Backend: ML model training. Frontend: Search & filters functional.

Week 6 (29 Mar - 4 Apr): Backend: ML API endpoint deployed. Frontend: Vehicle detail page.

Week 7 (5-11 Apr): Backend: Power BI dataset. Frontend: Price estimator integrated.

MILESTONE 2: ML model deployed, Search & Filter working, Estimator functional

Phase 3: Integration (Weeks 8-10: 12 Apr - 2 May)

Week 8 (12-18 Apr): Backend: API optimisation. Frontend: Power BI iFrame integration.

Week 9 (19-25 Apr): Backend: Data quality monitoring. Frontend: Market summary page.

Week 10 (26 Apr - 2 May): Backend: Load testing. Frontend: Cross-browser testing.

MILESTONE 3: Full integration complete, Power BI embedded, Platform functional

Phase 4: Launch (Weeks 11-14: 3 May - 30 May)

Week 11 (3-9 May): Bug fixes, UAT, documentation.

Week 12 (10-16 May): Deployment prep and analytics.

Week 13 (17-23 May): Production deployment and final polish.

Week 14 (24-30 May): Project presentation checklist, final testing, go-live prep.

MILESTONE 4: GO-LIVE 1 JUNE 2026

3. Backend Team Responsibilities

3.1 Damian | Backend Developer | Python Script, MySQL db. & Analytics

Layer 1: Data Pipeline

- Maintain Python scraper with error handling
- Schedule automated scrapes
- Monitor scraping logs and data quality
- Add support for new vehicle sites to be scraped

Layer 2: Database Management

- Optimise MySQL schema (indexes, query plans)
- Implement daily backups
- Create stored procedures for common queries
- Manage schema migrations and updates

Layer 3: ML & Analytics

- Design ML pipeline for price prediction
- Feature engineering (mileage/year, province encoding)
- Train Random Forest Regressor model
- Deploy ML model as REST API endpoint
- Retrain as the dataset expands

Coordination

- Guide Saturday checkpoint meetings
- Maintain project documentation
- Code review all backend PRs
- Manage Git branching strategy

Tools & Technologies

- Python 3.11+, Pandas, NumPy, Scikit-learn
- FastAPI (REST API)
- MySQL 8.x, MySQL Workbench
- PyCharm
- GitHub, pytest, Postman

3.2 Jude | Backend Developer | API & Login System

API Development

- Build REST API endpoints: GET /api/vehicles (paginated, filterable)
- GET /api/vehicles/:id (vehicle details)
- GET /api/search (full-text search with filters)
- GET /api/stats/summary (market overview)
- Implement authentication
- Write API documentation

Login & Access Controls

- Create User Management for access levels
- Password Security
- Role-Based Access Control

Data Quality

- Write data validation scripts
- Create data cleaning routines
- Monitor data consistency between scrapes

Tools & Technologies

- Python 3.11+, FastAPI
- Postman, Swagger
- MySQL, MySQL Workbench
- Power BI Desktop, Power BI Gateway
- VS Code

4. Frontend Team Responsibilities

4.1 Mpho | Frontend & Component Library

Project Setup (Week 1)

- React project
- Set up folder structure (based on different features)

Component Library (Weeks 1-3)

- Design reusable components: Button, Card, Input, Select, Modal, Table
- Vehicle card component with responsive design
- Pagination, LoadingSpinner elements
- Document components with examples
- Create theme system (colours, fonts, spacing)

Tools & Technologies

- React 18+
- Tailwind CSS
- VS Code with React extensions
- GitHub, Chrome DevTools

4.2 Taureen | Vehicle Listing & Search

Mock Data Setup (Week 1)

- Access mock vehicle data JSON (sample will be provided)

Vehicle Listing Page (Weeks 2-4)

- Build main listing page at /vehicles route
- Create responsive grid layout
- Add sorting dropdown (price, year, mileage)
- Implement pagination (20 vehicles per page)
- Add loading state pages

Search & Filter (Weeks 3-5)

- Full-text search bar
- Make/model dropdowns (multi-select)
- Price range slider, year range, mileage filter
- Transmission, fuel type, province, body type filters
- Show active filter chips with remove option\

API Integration (Week 4)

- Replace mock data with real API calls
- Handle loading states and errors

Vehicle Detail Page (Weeks 5-6)

- Build detail page
- Display all specifications in organised sections
- Implement image gallery
- Show similar vehicles carousel

Tools & Technologies

- React 18+, React Hook Form
- Tailwind CSS
- React Query
- VS Code, Chrome DevTools

Key Deliverables

- Week 3: Listing page with mock data
- Week 5: Full filter functionality with the real API
- Week 6: Vehicle detail page complete

4.3 Mpumelelo | UI & Power BI iFrame

Price Estimator Tool (Weeks 3-6)

- Build estimator page
- Design vehicle form: Vehicle basics, details, results
- Implement form validation with feedback
- Show comparison to market average

Power BI Integration (Weeks 6-8)

- Research Power BI embedding options
- Build dashboard page
- Implement secure iFrame embedding with Power BI API
- Configure responsive iFrame sizing
- Test across Chrome, Firefox, Safari, Edge

Market Summary Page (Weeks 7-9)

- Build summary page at /market-summary route
- Display key stats: total listings, average prices, trends

UX/UI Polish (Weeks 9-10)

- Ensure consistent design across pages
- Add smooth transitions and animations
- Implement error boundaries
- Create 404 and 500 error pages

Tools & Technologies

- React 18+
- Power BI JavaScript SDK
- Tailwind CSS, Framer Motion (animations)

Key Deliverables

- Week 6: Estimator working with ML API
- Week 7: Power BI iFrame embedded
- Week 8: Market summary page complete

5. Coordination & Saturday Stand-ups

Weekly Saturday meetings at 11:00 AM for 30 minutes for status and to align.

5.1 Saturday Meeting Agenda

- Updates (5 mins each): completed work, current tasks, blockers
- Milestone review: on track for this week's deliverables?
- Demo time: show working features
- Blocker resolution (15 mins max)
- Next week planning: confirm priorities

6. Deliverables Checklist

6.1 Backend Deliverables

- Python scraper running on schedule (daily)
- MySQL database with 70,000+ vehicle listings
- All API endpoints deployed and documented
- ML price prediction model deployed
- Power BI dataset published and refreshing
- Database backup active

6.2 Frontend Deliverables

- React app deployed to production
- Vehicle listing page with search, filters, pagination
- Vehicle detail page with specs and dealer info
- Price estimator tool integrated with ML
- Power BI dashboard embedded through iFrame
- Market summary page with charts
- Responsive design tested on different browsers
- Loading states, error handling implemented

6.3 Testing

- Tests for backend endpoints
- Integration tests for ML model
- Manual testing checklist completed
- User acceptance testing (UAT) with external testers
- All critical bugs fixed

7. Recommended Tools

7.1 Backend Team Tools

- PyCharm and VS Code (Python IDE)
- MySQL Workbench (database)
- Postman (API testing)
- GitHub (code repo)

7.2 Frontend Team Tools

- VS Code (React support)
- Chrome DevTools (debugging, responsive testing)
- React DevTools (component inspection)
- Figma (design inspiration/collab)

8. Learning Resources

React (Frontend Team)

- Official React Docs: <https://react.dev/learn>
- React Course (free): <https://scrimba.com/learn/learnreact>
- React Tutorial (YouTube)

FastAPI (Backend Team)

- FastAPI Tutorial: <https://fastapi.tiangolo.com/tutorial/>
- Build a REST API in 30 minutes (YouTube)

GitHub

- GitHub Learning Lab: <https://lab.github.com/>